10-617: Project Proposal

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1 Problem Statement

CLIP pairs a vision and a text encoder in a shared space. Prior work compresses it but reports only final accuracy at a fixed compression level.

We ask: **as CLIP is compressed toward failure, which modality's alignment collapses first, and at what ratio?** We hypothesize this order is predictable from the uncompressed model's geometry, with *alignment* and *uniformity* as candidate predictors. Contributions: a per-modality degradation-ordering method and a geometric predictor of which modality fails first.

2 Methods of Evaluation

Model/compression. We compress a frozen pretrained OpenCLIP ViT-B/32, primarily via structured pruning applied per encoder (a continuous ratio sweep, single GPU with short recovery fine-tuning), with post-training quantization as a secondary axis.

Data (standard/public via OpenCLIP/HF). Zero-shot classification (ImageNet-1k) and image–text retrieval (MS-COCO, Flickr30k).

Probes & predictor. Per compression level we run separate vision and text probes, compressing one encoder at a time (against the other at full precision) to attribute collapse; collapse ratio = where a probe drops a fixed margin below full precision. We then test whether geometric measures (alignment, uniformity, effective rank) on the full model predict the collapse order and ratio.

3 Division of Labour, Timeline

Labour. Akshu: compression pipeline and per-encoder controls. Nishnath: probes, geometric metrics, and predictor analysis. Shared: design and writing.

Timeline. W1–2: OpenCLIP + data, baselines, pruning sweep with both probes. W3–4: single-encoder controls, geometric metrics, predictor test. W5–6: quantization axis / second backbone, and final report.